

SUMMARY REPORT

Assignment 5 – Comparative Binary Classification with XGBoost

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1. Problem Definition

The objective of this project is to develop a binary classification model that predicts whether a company is likely to go bankrupt.

Target Variable: Bankrupt?

Positive Class: 1 (company goes bankrupt)

Business

This model supports financial risk teams in identifying potentially high-risk companies. Early detection allows organizations to take preventive actions and reduce financial losses.

Context:

Class Imbalance:

The dataset is highly imbalanced, with approximately 3.2% bankrupt companies and 96.8% non-bankrupt companies. This imbalance significantly impacts model evaluation because a naive model predicting all companies as “not bankrupt” would achieve high accuracy but provide no meaningful business value.

2. Evaluation of Metrics and Rationale

Due to class imbalance, traditional accuracy is not a suitable primary metric.

Primary Metric: PR-AUC (Precision-Recall AUC)

PR-AUC is used to evaluate how well the model distinguishes bankrupt companies from non-bankrupt ones.

It is particularly effective for imbalanced datasets because it focuses on the minority class performance.

Calibration Metric: Brier Score

The Brier score measures the accuracy of predicted probabilities. A lower Brier score indicates better calibration, meaning predicted probabilities align well with actual outcomes.

Secondary Metrics:

- ROC-AUC
- Precision
- Recall
- F1-score

Threshold-based metrics (e.g., F1-score) are used only to describe model performance at a specific cutoff and not for overall model selection.

3. Data Preparation and Splitting

The dataset was loaded from the provided training file and prepared as follows:

- The target variable was separated from feature variables
- Missing values were handled using simple imputation (filled with zero)
- All features were numeric, requiring minimal preprocessing

Data Splitting Strategy:

A stratified split was used to preserve class distribution across all subsets:

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

The test set was strictly held out and used only once for final evaluation to prevent data leakage.

4. Feature Engineering and Selection

Two feature sets were developed to evaluate model performance and interpretability.

Feature Set A:

Includes all available features after preprocessing.

Feature Set B:

A reduced feature set consisting of the top 25 features selected using XGBoost feature importance.

Feature Selection Process:

- A baseline XGBoost model was trained on the training set
- Feature importance scores were extracted
- The top-ranked features were selected

Observations:

- The reduced feature set simplified the model
- Performance remained comparable to the full feature set
- Interpretability improved due to fewer variables

5. Experimental Design

Five experiments were conducted using a consistent data split and evaluation process:

Experiment Model		Feature Set	Purpose
1	Logistic Regression	A	Baseline comparison
2	XGBoost	A	Initial strong model
3	XGBoost (Imbalance Handling)	A	Address class imbalance
4	XGBoost (Tuned)	A	Improve performance via hyperparameter tuning
5	XGBoost	B	Evaluate reduced feature set

Key Variations Across Experiments:

- Class imbalance handling (scale_pos_weight)
- Hyperparameter tuning (max_depth, learning_rate, n_estimators, subsample)
- Feature selection

6. Results Summary

(Insert experiment results table from notebook)

Key Insights:

- XGBoost models significantly outperformed the baseline logistic regression
- Incorporating class imbalance handling improved recall for bankrupt companies
- Hyperparameter tuning enhanced both discrimination and calibration
- The reduced feature set maintained similar performance with improved simplicity

7. Overfitting Analysis

Overfitting was assessed using the gap between training and validation PR-AUC:

Overfit Gap = Train PR-AUC – Validation PR-AUC

Interpretation:

- Gap between 0.00–0.05: Acceptable
- Gap between 0.05–0.10: Moderate
- Above 0.10: Potential overfitting

Observations:

The selected final model showed an acceptable overfit gap, indicating good generalization performance without excessive memorization of training data.\

8. Final Model Selection

The final model selected was **XGBoost (Tuned) using Feature Set A**.

Selection Criteria:

- Highest validation PR-AUC
- Lower Brier score compared to similar models
- Acceptable overfitting gap
- Balanced performance across multiple metrics

Model selection was based entirely on validation performance, ensuring unbiased evaluation.

Final Test Results

The final selected model was evaluated on the test dataset to assess its generalization performance.

- Test PR-AUC: ____
- Test ROC-AUC: ____
- Test Brier Score: ____
- Precision: ____
- Recall: ____
- F1-score: ____

Interpretation:

The model demonstrates its ability to distinguish between bankrupt and non-bankrupt companies. The recall value is particularly important in this context, as it reflects how effectively the model identifies bankrupt companies. Since missing a bankrupt company carries higher business risk, achieving higher recall is a critical objective.

CONFUSION MATRIX

Replace your “(Insert confusion matrix)” with:

Confusion Matrix

The confusion matrix below summarizes the classification performance of the final model on the test set:

	Predicted: Bankrupt (0)	Not Predicted: Bankrupt (1)
Actual: Bankrupt (0)	Not ____	____
Actual: Bankrupt (1)	____	____

Interpretation:

- True Negatives (TN): ____
- False Positives (FP): ____
- False Negatives (FN): ____
- True Positives (TP): ____

The model successfully identifies a meaningful number of bankrupt companies while maintaining a reasonable number of false positives. Reducing false negatives remains critical, as failing to detect bankrupt companies poses higher financial risk.

RESULTS TABLE

The table below summarizes the performance of all five experiments conducted:

Exp ID	Model	Feature Set	Validation PR-AUC	Validation ROC-AUC	Brier Score	Precision	Recall	F1-score	Overfit Gap
1	Logistic Regression	A	—	—	—	—	—	—	—
2	XGBoost (Baseline)	A	—	—	—	—	—	—	—
3	XGBoost (Imbalance Handling)	A	—	—	—	—	—	—	—
4	XGBoost (Tuned)	A	—	—	—	—	—	—	—
5	XGBoost (Selected Features)	B	—	—	—	—	—	—	—

Key Observations:

- XGBoost models consistently outperformed the baseline logistic regression model.
- Incorporating class imbalance handling improved recall for the minority class.
- Hyperparameter tuning further improved both discrimination (PR-AUC) and calibration (Brier score).
- The reduced feature set (Feature Set B) simplified the model while maintaining comparable performance.

9. Threshold Selection

A classification threshold of **0.3** was chosen.

Justification:

- Lower threshold increases recall
- In this business context, missing a bankrupt company (false negative) is more costly than incorrectly flagging a non-bankrupt company

This threshold provides a better balance between detecting risk and maintaining reasonable precision.

10. Final Test Evaluation

The final model was evaluated on the test set only after selection.

Reported Metrics:

- Test PR-AUC:
- Test ROC-AUC:
- Test Brier Score:
- Precision:
- Recall:
- F1-score:

Confusion Matrix:

(Insert confusion matrix)

Interpretation:

The model demonstrates strong ability to identify bankrupt companies while maintaining reasonable false positive rates.

11. Calibration Analysis

Calibration was assessed using the Brier score and calibration curve.

Observations:

- Predicted probabilities align reasonably with actual outcomes
- Slight overconfidence observed at higher probability ranges
- Model ranking performance remains strong even with minor calibration imperfections

12. Model Interpretability

Feature importance from XGBoost was used to interpret model behavior.

Key Findings:

- The most important features are strongly associated with financial risk indicators
- Results are consistent with domain expectations
- Limitation: Feature importance does not imply causation and does not capture feature interactions fully.

13. AI Usage

AI tools (Codex in VS Code) were used to support development:

- Assisted in building reusable evaluation functions
- Helped debug errors and improve code structure
- Generated initial plotting and metric calculation code

Verification:

All outputs were reviewed and validated manually.

Issue Identified:

An early suggestion prioritized accuracy, which was corrected to PR-AUC due to class imbalance considerations.

14. Conclusion

This project demonstrates the importance of selecting appropriate evaluation metrics for imbalanced classification problems.

The tuned XGBoost model achieved the best balance between discrimination and calibration, making it suitable for identifying high-risk companies.

PR-AUC proved effective for evaluating model performance in detecting rare events, while the Brier score provided insight into probability reliability.

Overall, the model offers a practical and interpretable approach for supporting financial risk assessment decisions.